BASHNAV

One icon, all the tools behind it

Version 1.0 — the launcher: a list at a prompt, and a page in a browser. Six tools for a boat with no signal.

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1. What this is

BASHNAV is not a tool in its own right. It is the way in to the others:

celnav	celestial navigation: sight reduction and the fix
colregs	the rules of the road, lights, and collision avoidance
tides	tide prediction for 8,334 stations
deck-log	the boat's records: deck, engine, provisions
weather	read your own barometer
drill	a study machine, browser build only

It exists in two forms, and both of them are a list: one you read at a prompt, one you tap.

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2. Why a list rather than five icons

On an iPad each tool *can* have its own home-screen icon through a Shortcut — but that needs the Shortcuts action set to run **in app** and not in the extension, because the extension has no terminal and an interactive menu has nothing to talk to. That is a setting people get wrong once and cannot diagnose.

The launcher is the version that always works. It is also the honest answer on a boat: **you are cold, the screen is wet, and finding the right icon among five is worse than reading a list of five.**

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3. At a prompt

Run `bashnav`. It prints the list, you pick a number, it runs that tool and returns you to the list when the tool exits.

How it finds the tools, in this order, because a launcher that cannot find them is worse than no launcher:

1. next to the launcher script itself;
2. a `bin/` directory beside it;
3. `~/Documents` — which is the place iOS actually lets an app write;
4. `~/bin`;
5. the current directory;

6. anything on your `PATH`.

A tool that is not found is shown as missing rather than failing when you pick it.

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4. In a browser

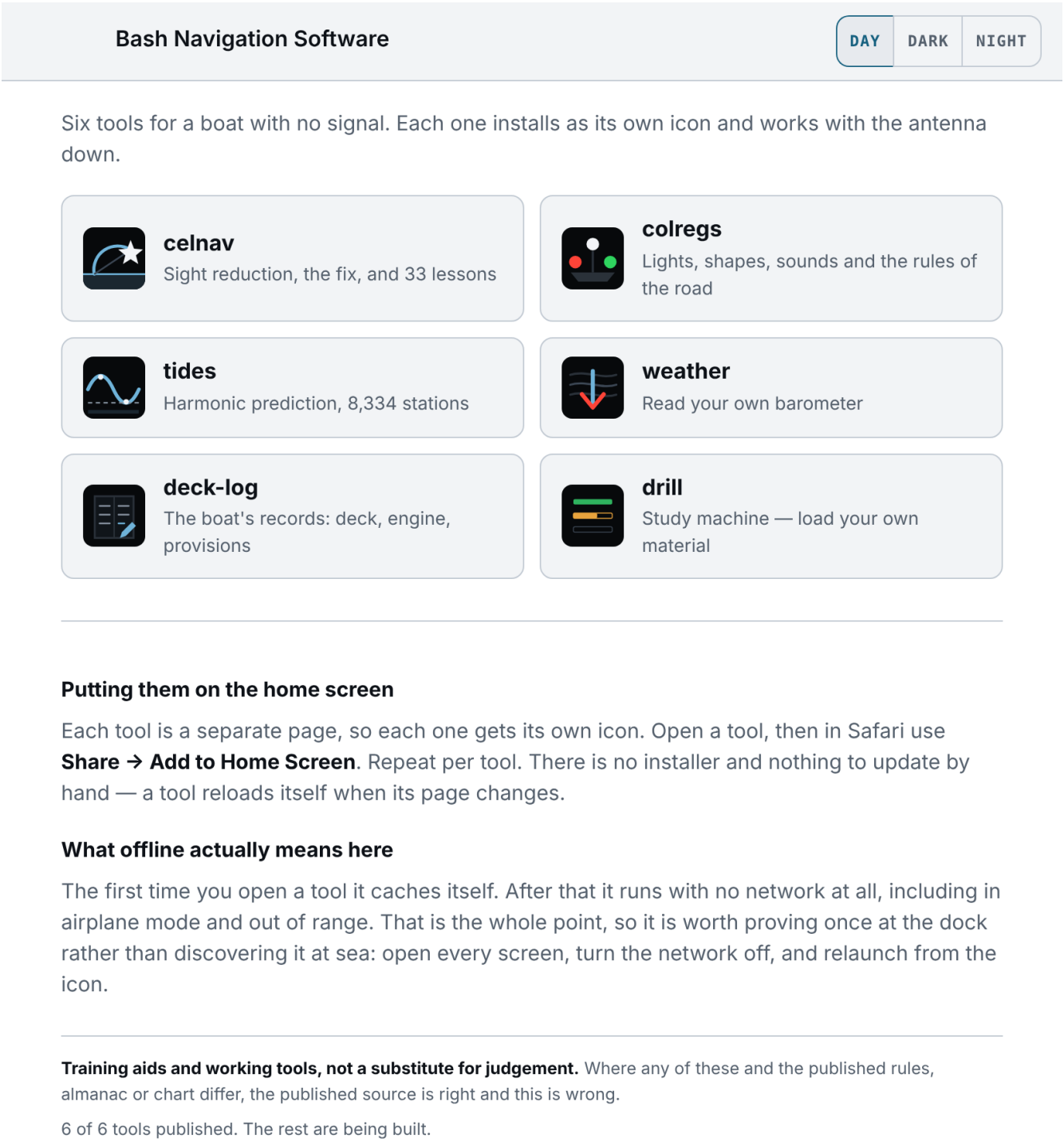


Figure 1 — the launcher page: six tools, the three display modes, and what to do once.

Each tool is a separate page, so each one gets its own icon. There is no installer and nothing to update by hand: a tool reloads itself when its page changes.

1. Open a tool.

2. In Safari, **Share → Add to Home Screen**.
3. Repeat for each tool you want on the screen.

What offline actually means here

The first time you open a tool it caches itself. After that it runs with **no network at all** — airplane mode, out of range, antenna down.

Prove it once at the dock rather than discovering it at sea: open every screen of the tool, turn the network off, close it completely, and relaunch it from the icon. A tool that was only ever opened online has never shown you that it works offline; it has only shown you that the network works.

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5. The display modes

The launcher page carries the same three modes as the tools, and they share one setting across the whole suite.

The display: day, dark and night

The browser build has three display modes. They are not decoration: one of them exists to protect your night vision, and choosing the wrong one at the wrong moment costs you something real.

Mode	What it is for
NIGHT	Red on black, for a dark-adapted eye. The default.
DARK	Near-black with light text, for below decks, dusk, or indoors.
DAY	White with near-black text, for daylight and for printing.

Why night is the default

A tool you open on deck in the dark must not start bright. A bright panel costs a dark-adapted eye roughly **twenty minutes** to recover from, and twenty minutes is a long time to be unable to see a mark or a ship’s lights properly. So the first time you open any of these tools you get night, and from then on you get whatever you last chose.

Three buttons, not a cycle

The three modes are three separate buttons, and the one you are in is shown as selected. That is deliberate.

1. **A cycle can take you somewhere you did not intend.** One stray press with a cycling control can land you on DAY at three in the morning.
2. **A cycle cannot tell you where you are**, only what the next press would do. Three buttons show the current mode at a glance without changing it.
3. **Going darker is always one tap.** DARK and NIGHT are immediate from anywhere, because going darker never costs you anything.

4. **Going from NIGHT to DAY takes a deliberate half-second press.** Hold the DAY button and it fills as it goes; let go early and nothing happens. Nothing bright appears on the screen until the change actually completes. That one transition is guarded because that one transition is the only one that costs you your dark adaptation.

The setting follows you

The mode is remembered for all of the tools together, and for the course pages that embed them. Set night in one and you are in night in the next. It is remembered in the browser, so it survives closing the app.

It is right before the page draws

The mode is set in the first instructions the page runs, before any of it is painted. This matters on a first visit over a poor connection: set it later and the page would appear in one palette and then jump to another, which at night is the difference between keeping your night vision and losing twenty minutes of it. If nothing has been stored yet, or whatever was stored cannot be read, the page opens in **night**.

The controls follow the palette too

Sliders, scroll bars and the other controls the browser draws are told which world they are in, so they are drawn in the mode you chose rather than the browser's own light default. Without that, a night screen can carry a bright white slider across the middle of it.

If you use High Contrast Mode

On Windows, High Contrast replaces every colour on the page with the ones your system theme specifies. That applies here too: the tools do not override it, because it is your setting and you chose it.

Nothing is lost by it. Where a colour carries meaning — the navigation lamps in colregs — the letter beside it says the same thing, so a lamp is still identifiable when the hue has gone.

One thing to know if you need both. High Contrast wins over night mode. If you need night vision on deck *and* high contrast to read, choose a dark high-contrast theme in Windows; the tools will follow it.

What the display can and cannot do in sunlight

Worth knowing before you blame the app on a bright day.

In sunlight, light reflecting off the glass lands on the dark parts of the screen and the light parts alike. That lifts everything together and flattens the difference between them, so the contrast you actually see collapses no matter what colours the page uses:

Condition	Body text, as seen
Indoors	about 10:1
Overcast on deck	about 6:1
Bright	about 3.5:1
Full sun on the glass	about 2.3:1

At the extreme, every colour scheme converges. Pure black on pure white measures the same 2.3:1 in full sun as the day mode does, because the glare swamps the difference. A different palette is not the answer, and nor is a different shade of white: across every plausible choice of background the contrast changes by under 4%, which is below what an eye can see.

What does work, in order:

1. **Turn the screen brightness up.** This is the one large lever, and it is the device's, not the app's.
2. **Shade the screen** with a hand, your body or the sprayhood. Removing the reflection is worth more than any change the app can make.
3. **Use DAY mode**, which at least puts the maximum luminance the panel can give behind the darkest text it can draw.

A note for printing

Documents and their figures are made in **day** mode. A screen captured in dark or night mode prints as a solid block of toner and loses the fine type in a working, so the printed pages show the day display even if that is not the one you use at sea.

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6. Cautions

1. **Training aids and working tools, not a substitute for judgement.** Where any of these and the published rules, almanac or chart differ, the published source is right and this is wrong.
2. **The browser tools keep their data on the device they run on.** Nothing is uploaded and nothing is synchronised. Export anything you would mind losing.
3. **Prove offline before you need it.** See section 4; it takes five minutes alongside and cannot be done at sea.